

PM Case Challenge Aplazo

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Problem Statement

GMV is flat in the last 2 quarters, the reasons are unknown.

Riskiest Assumption: the affected reason is shared on all merchants somehow, this is quite risky as it changes all the strategy to attack the problem because of the impact.

Experiment: Compare the estimated GMV opportunity across merchant GMV tiers and test whether the top 20% of merchants account for a disproportionate share of the opportunity

Evidence: If it results positive the 20% of merchants as a disproportionate share, i would reject the hypothesis and prioritize a targeted solution.



Data Analysis

funnel_step_reason		COUNT	de ses
[- completed		28637	
	confirmation	192	
	declined	216	
	kyc	214	
Total completed		29259	
[- confirmation	confirmation	2815	
Total confirmation		2815	
[- kyc	kyc	5189	
Total kyc		5189	
[- underwriting	declined	1909	
Total underwriting		1909	
[- widget_interacted	payment_method	24067	
Total widget_interacted		24067	
[- widget_shown	widget_no_interacted	65675	
Total widget_shown		65675	
Suma total		128914	

Drop off reasons (only 128,914 widget shown):

Widget not interaction: 65,675 (50.9%)

Payment not selected: 24,067 (18.7%)

Other reasons (KYC, underwriting and confirmation): 9,913 (7.7%)

29,259 completed checkout (22.7%)



Data Issues/Inconsistencies

- The widget funnel step is reached but the widget is not always showing and much less is interacted with, so how do we know if it's really reached or if there is a bug? Is this funnel step really needed? for this exercise i excluded this flag from funnel calculations.
- 27 merchants have `sandbox_passed` after `production_credentials` so this can have an inconsistency on the data and the expected flow.
- There are merchants that have `went_live` on the integration but are flagged as churned, this can be normal but we need to implement the churn date flag to review if we need to disconnect or to review if there are checkout sessions after this date that affect the GMV.
- Platform names are not homologated, this can cause a data miss interpretation because you will analyze 12 platforms instead of 4.
- There are 622 sessions that have a drop-off reason but they are marked as completed.



Top Insight

WooCommerce has a conversion issue on mobile, it has an interaction of 27.1% and a completion rate of 12.2% vs desktop that it has 54% of interactions and completion of 26.5%.

COUNTA de session_id		funnel_step_reached						
platform	device	completed	confirmation	kyc	underwriting	widget_interacte	widget_shown	Suma total
Custom API	desktop	26.17%	2.63%	2.92%	2.07%	20.11%	46.10%	100.00%
	mobile	22.37%	2.36%	5.17%	1.60%	19.08%	49.43%	100.00%
Total Custom API		23.54%	2.44%	4.47%	1.75%	19.40%	48.40%	100.00%
Shopify	desktop	26.35%	2.71%	2.67%	1.71%	20.48%	46.08%	100.00%
	mobile	22.92%	2.07%	4.95%	1.46%	19.54%	49.06%	100.00%
Total Shopify		23.99%	2.27%	4.24%	1.53%	19.83%	48.13%	100.00%
VTEX	desktop	26.21%	2.69%	2.86%	1.68%	19.94%	46.63%	100.00%
	mobile	23.30%	2.15%	4.96%	1.24%	19.55%	48.80%	100.00%
Total VTEX		24.20%	2.32%	4.31%	1.37%	19.67%	48.12%	100.00%
WooCommer	desktop	26.47%	2.41%	2.63%	1.59%	20.90%	45.99%	100.00%
	mobile	12.21%	0.98%	2.45%	0.89%	10.52%	72.95%	100.00%
Total WooCommerce		16.59%	1.42%	2.51%	1.11%	13.72%	64.66%	100.00%
Suma total		22.70%	2.18%	4.03%	1.48%	18.67%	50.94%	100.00%



System Rank

Business Impact: we take in count the tier GMV, GMV relative loss and GMV total relative to calculate it.

Severity: We take in count the average stage where the users drop off, non-completion rate, widget not shown and integration errors to give the recommendations.

Confidence: We take in count the volume of sessions and merchant status.

Scalability: We take in count how many merchants have almost a similar issue based on platform, integration, device and drop-off reason.



Business Case

- Listen and understanding of the other initiative.
- Argument why my initiative is more relevant vs the other one.
- If the solution takes 3 months of development:
 - Understand why is this time needed.
 - Propose to phase it if possible.
 - If even the first phase/MVP needs 3 months, re-evaluate it with potential outcome vs opportunity cost of the other initiatives.

NOTE: If during development i notice that our diagnostic was not clear enough or has an opportunity during the analysis i would stop it as the evidence is clear enough for it and give the backlog space for another initiative



Attractive but not pushable

High GMV value but low scalability, this can sound awesome when you mention the \$\$ but if it's only adaptable to some merchants it will be risky.

Custom merchants, this also can sound attractive but as the name says it they have custom plugins so it may only apply to that specific merchant unless the GMV opportunity is large enough to justify its lack of scalability.

Merchants that have high ranking but have been decreasing their sessions lately, this is because maybe the merchants are near to a churn so first we should align with them in the near future before any development starts.



Defend it under pressure

Engineering: I would like to understand the reasons you found so we can discuss if you are right, my tool has taken multiple variables that points that this initiative should take the first priority but if i missed something that you noticed i would like to know it so i can adjust it and validate the new priority that is needed.

KAMs: We can prioritize the merchant that you want, at the end the decision is yours, but i want to emphasize that the tool has multiple variables behind that point that the merchant we selected is the most optimal to achieve a better business impact.

Leadership: Is there a reason that I don't acknowledge that supports this ask; is the merchant selected going to churn or disconnect? The tool has the capacity to change the prioritization balance between GMV and scalability so we can move this and select a different strategy.

I'm flexible on the solution, scope and sequencing but, I would not be flexible on the change of any of this without evidence that justifies it.



Roadmap

30 days: Feed the tool to demonstrate that it is well configured and ready to give realistic recommendations.

- Homologate the data to reduce risks in the analysis.
- Complete the data with more variables such as churn date or disconnection date to increase the confidence of the tool

Next Quarter: The tool becomes a trusted input into merchant prioritization

- Automate the data feed with the DATA team.
- Let the KAMs input their comments, suggestions and impact of actions so the tool can contrast the real impact vs the estimate it did and adjust for the next initiatives proposed.

Next 6 months: The tool has demonstrated that it takes the best decisions and is a complement for KAMs instead of a competition

- Automate the KAMs inputs and impact measures.
- Adjust/increase the ranking variables because the market moves constantly and the tool needs to adapt



Adoption

- KAM involvement since initiative raising
- Change management

Measurement::

- Decisions taken with tool information (KAM input).
- Business impact done after a decision taken from tool recommendation.
- Override rate + quality, if the KAM decides to go with his own decision and it's better we should measure this as this will help the system to learn for a future.
- I would **not use** log on KAM login or clicks as they can only login and navigate through it but without taking decisions with the tool.



Observability

Contrast information from the past:

1. Input data from the past
2. Request prioritizations for the tool and contrast it to the one that the KAM did in that time before the tool existed and compare results with:
 - a. Potential GMV captured vs real GMV obtained with KAM decision.
 - b. Incremental GMV, how much GMV of difference is between the potential decision of the tool and the one the KAM decided.
 - c. Conversion incremental vs real conversion incremental.
 - d. Gaps that KAM didn't identify for any kind of reason.

If this meets all the conditions then we can be sure that the tool is working properly.



Failure Modes

- High GMV merchants prioritization, if even when the tool we have bias for the biggest merchants we have a problem; we need to adjust the scalability.
- Low confidence prioritization, if the tools rank higher merchants that have contradictory data we have an issue; we will need to adjust the confidence variable.
- When we introduce small samples of data they rank top, this will be a failure because it shows that the model overreacts; install a rule that for a merchant to enter the ranking you will need a minimum of sessions.
- Lifecycle fail, if any churned merchant appears on any date after churned its a failure reason; flag the merchants churned date and alert if any data is corrupted.



Summary

The goal is not to build a smarter dashboard. Is to implement a decision supported system that gets better overtime with more data and with KAM (users) feedback. It will only be successful if the data after the evaluation period is concrete on the results, that it's demonstrated that it generates better outcomes and business impacts than human judgement.